

Qwerty: A Basis-Oriented Quantum Programming Language

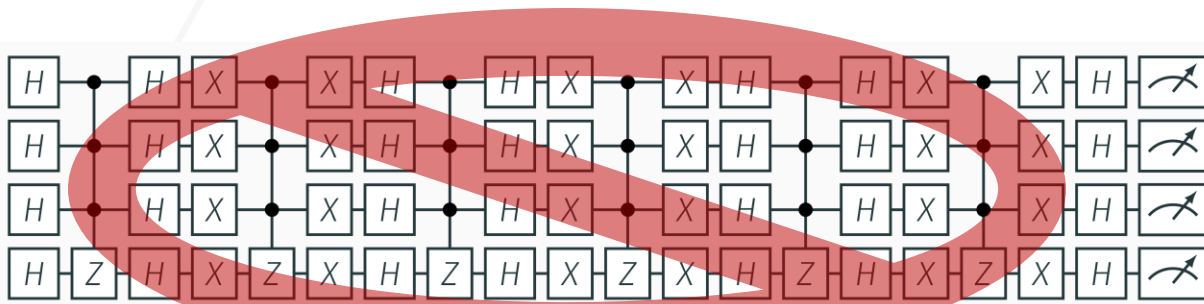
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<https://qwerty.cc.gatech.edu/>

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Quantum computers promise exponential speedups,
but quantum programming is painful!



Quantum circuit (prior languages)

Tedious, like writing classical assembly



'p' ** N >> - 'p' ** N

Qwerty (our language) [QCE '25]



Successful & expressive

Our compiler compiles Qwerty to circuits [CGO '25]

Written in: Rust, C++ (MLIR, LLVM), Python

Rough Project Ideas

1. Increase test coverage from ~35% to 80%
2. Beta test and improve error messages
3. Upload Qwerty package to PyPI
4. Make Qwerty interpreter more robust
5. Rewrite type inference
6. Optimize classical to quantum synthesis

**Our open-source Qwerty compiler synthesizes efficient
circuits from high-level quantum code**

Source Code:



github.com/gt-tinker/qwerty

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